



Multiple-image encryption algorithm based on a new composite chaotic system and 3D coordinate matrix

Qisu Gao, Xiaoqiang Zhang^{*}

School of Information and Control Engineering, China University of Mining and Technology, Xuzhou 221116, China

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ABSTRACT

The image encryption technology is an important means to ensure network digital data transmission security. To improve the security and encryption computational efficiency, this paper proposes a new chaos-based image encryption algorithm. First, a new two-dimensional chaotic system is designed by coupling Sine and Henon maps. Comparative tests with other chaotic maps are conducted, and the results show that the new hybrid chaotic map has better chaotic performance in terms of the bifurcation diagram, Lyapunov exponent, sensitivity, etc.; second, a three-dimensional coordinate matrix is designed by the new chaotic system for the scrambling operation of multiple images; finally, the exclusive OR operation based on chaotic system is used for the pixel diffusion to obtain encrypted images. Experimental results show that the histograms of the encrypted images are even and the correlation coefficients approach 0. Meanwhile, the proposed algorithm can effectively resist the chosen plaintext attack, brute force attack, clipping attack, salt and pepper noise attack, etc.

1. Introduction

With the development of Internet technology, huge amounts of data are transmitted over the Internet. Digital images carry a large amount of visual information, and their transmission security has attracted widespread attention [1]. However, the digital image has substantial data redundancy and relevance, and traditional text encryption methods do not apply to the digital image. To protect image information, it is particularly important to design an effective image encryption algorithm.

Chaos is an irregular disordered state. The properties of chaotic maps, such as extreme sensitivity to initial conditions, non-periodicity to motion trajectories, unpredictability, and nonlinearity, make them suitable for image encryption [2]. Chaotic maps can be distinguished as one-dimensional (1D) and multidimensional (MD) maps. 1D chaotic maps have a few parameters, simple trajectory and short generation time [3]. MD chaotic maps have more parameters and more complex trajectories which makes them much harder to predict their behavior [4]. Due to the more complex structure, MD chaotic maps take longer time to generate chaotic sequences. In recent years, many scholars have improved traditional chaotic maps. Wang et al. [5] proposed an image encryption based on an efficient 1D composite chaotic map. The new chaotic map improves the traditional Logistic map and increases the

range of chaos initial values. Yousif et al. [7] used four-dimensional (4D) chaotic system to scramble and diffuse the image, making the encryption algorithm have a larger key space. The five-dimensional (5D) chaotic system constructed by Ning et al. [8] has a hyper-chaotic effect and can effectively resist various attacks. Therefore, designing a chaotic map with better chaotic performance is crucial.

Many scholars couple the original chaotic maps to construct a new chaotic map to improve the performance of chaos. For example, Hu et al. [37] coupled Logistic map and Sine map to improve the bifurcation diagram of chaotic map. Hua et al. [30] combined the Logistic map and the Sine map to construct a new two-dimensional (2D) chaotic map which has better key sensitivity. An improved 2D chaotic map is proposed by Gao et al. [38] based on the Sine and Tent maps. Wu et al. [39] proposed a 2D chaotic map named the 2D Henon-Sine map. The phase diagram of this chaotic map is improved. It can be seen that the method of coupling chaotic maps can indeed improve the performance of chaotic map. Regardless of the method used to construct chaotic map, scholars hope to obtain a better pseudo-random sequence for using in image encryption. The proposed chaotic map provides better randomness in bifurcation diagram and a longer chaotic range than the traditional chaotic maps.

Image scrambling operation means changing the pixel position and scholars have proposed many different image scrambling methods. The

^{*} Corresponding author.

E-mail address: grayqiang@163.com (X. Zhang).

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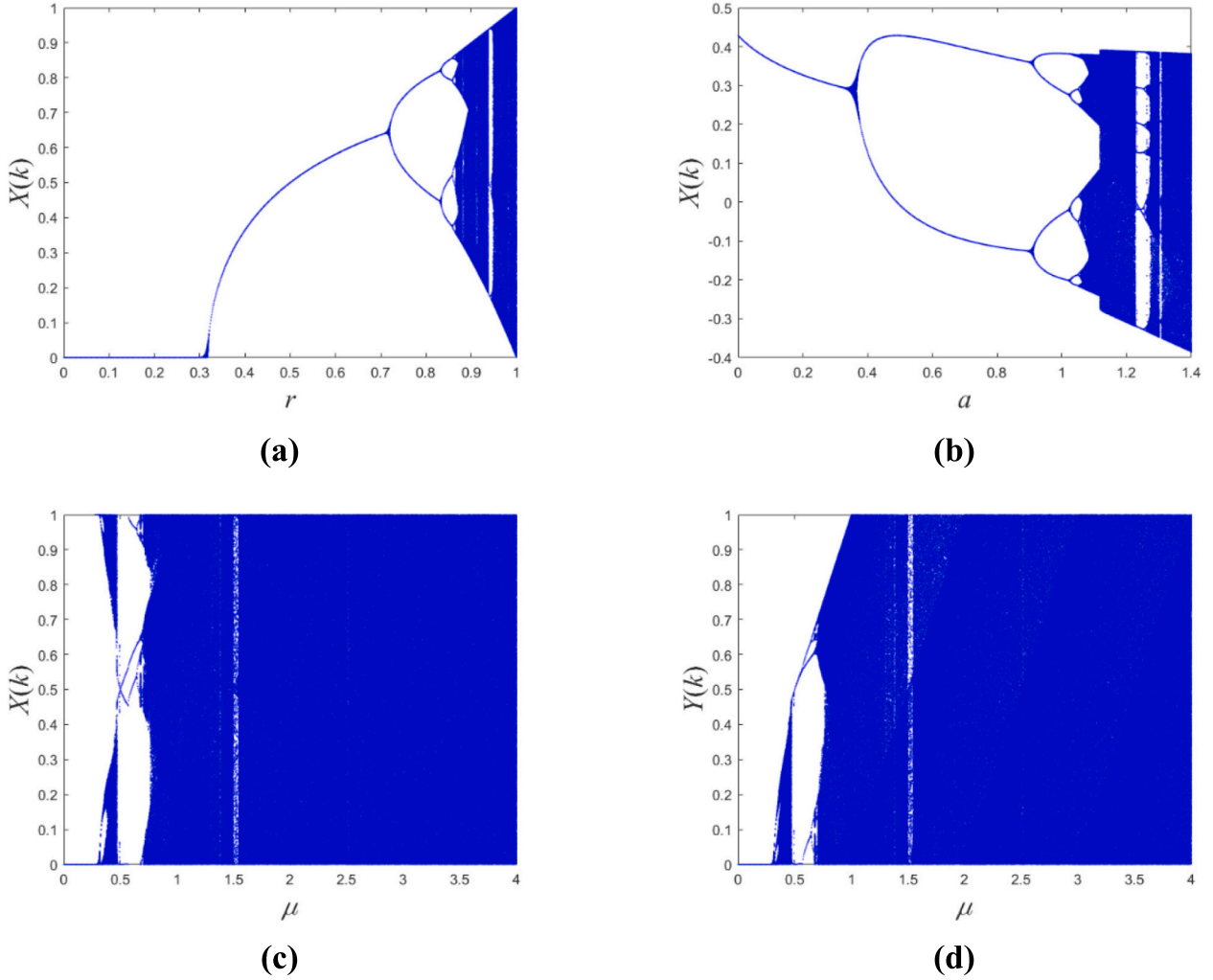


Fig. 1. Bifurcation diagrams. (a) Sine map, (b) output X of Henon map with $b = 0.3$, (c) output X of the 2DSHM, (d) output Y of the 2DSHM.

traditional Arnold transformation uses the Arnold matrix to change the pixel position. Tora et al. [9] proposed an improved Arnold transform. This algorithm expands the original parameter matrix of Arnold transform, iterates the Arnold transform a random number of times on the image, and reduces the periodic effect of the Arnold transform. Delavar et al. [10] proposed a new chaos-based image encryption algorithm. This algorithm combines the Arnold transformation and Logistic map to generate chaotic sequences. Meanwhile, the image is scrambled in blocks. Although the Arnold transformation can effectively change the pixel position, the encryption computational efficiency is low. Meanwhile, the image size is limited, and it is only used to encrypt a single image.

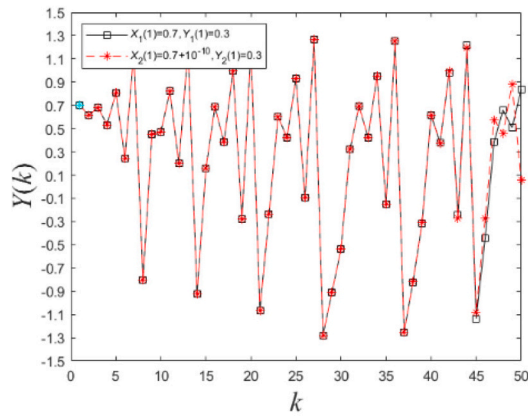
Zigzag transformation disrupts the position of pixels through a fixed traversal method. Guo et al. [11] designed an improved reverse Zigzag transformation. This algorithm improves Zigzag traversal method and reduces the sequence correlation after scrambling. Li et al. [12] applied the Zigzag transform at the bit level to scramble the pixel position and change the pixel value. Although the Zigzag transformation is simple, the positions of the first and last elements have not changed before and after the transformation. Some scholars propose the improved Zigzag transform, and experimental results and security analyses show that the improved Zigzag transform has good encryption effects and can resist common attacks. However, the Zigzag transform is limited to the 2D space and the scanning starting point is limited. These scrambling operations without chaotic maps still have high correlations in the

encrypted image after a single scrambling. The encrypted images of the encryption algorithms that use chaotic maps to descramble the images will often have lower correlations.

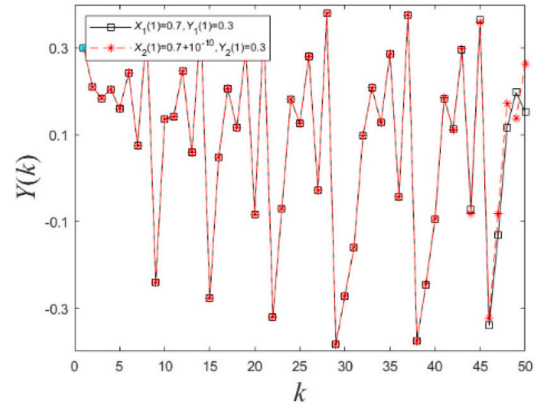
Multiple-image encryption algorithms can encrypt multiple images at the same time. Demirtas [6] proposed a multiple-image encryption which can encrypt both color and grayscale images. However, the size of images encrypted by this algorithm must be consistent. He et al. [13] use chaotic map they proposed and discrete wavelet transform (DCT) to encrypt multiple images. The algorithm compresses the images to reduce encryption time. However, since the DCT is a lossy compression, the decrypted image cannot be 100 % consistent with the original images. Kumar et al. [14] divided the images into three encrypted images and used the DNA algorithm to change the pixel values. However, the correlation coefficients of the encrypted images are higher than that of some existing algorithms. Some existing multiple-image encryption algorithms have the problems of fixed encrypted image size and low security. To solve these problems, a new multiple-image encryption algorithm is proposed in this paper.

This paper constructs a new 2D chaotic map and a new three-dimensional (3D) scrambling method. The main contributions of this paper are listed as follows.

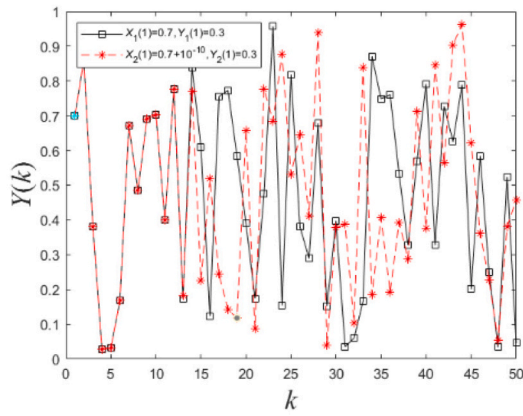
- (1) The 2D Sine-Henon map called 2DSHM is constructed. It is generated by a coupling operation between 1D Sine and 2D Henon maps. Compared with some existing chaotic maps, the



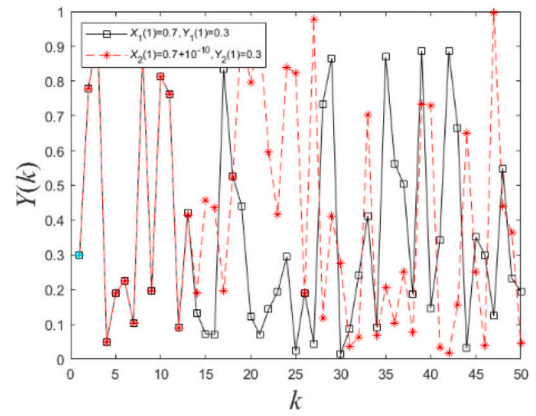
(a)



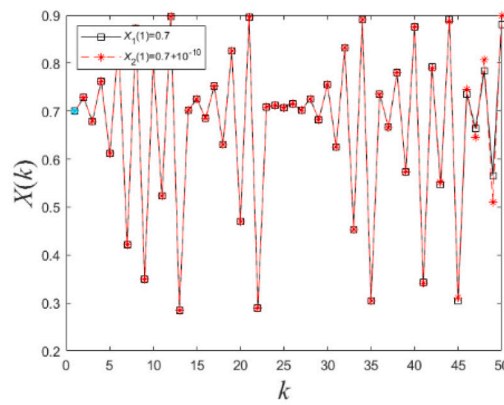
(b)



(c)



(d)



(e)

Fig. 2. Trajectory diagrams. (a) Output X of Henon map, (b) output Y of Henon map, (c) output X of the 2DLHM, (d) output Y of the 2DLHM, (e) output of Sine map.

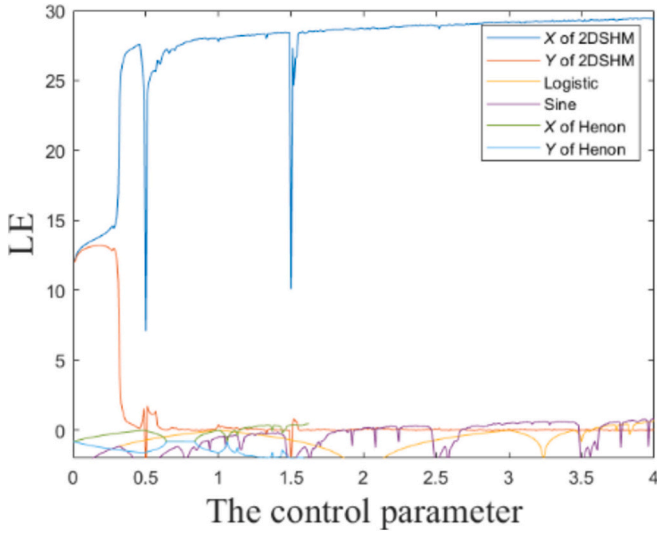


Fig. 3. Comparisons of the LEs for four chaotic maps.

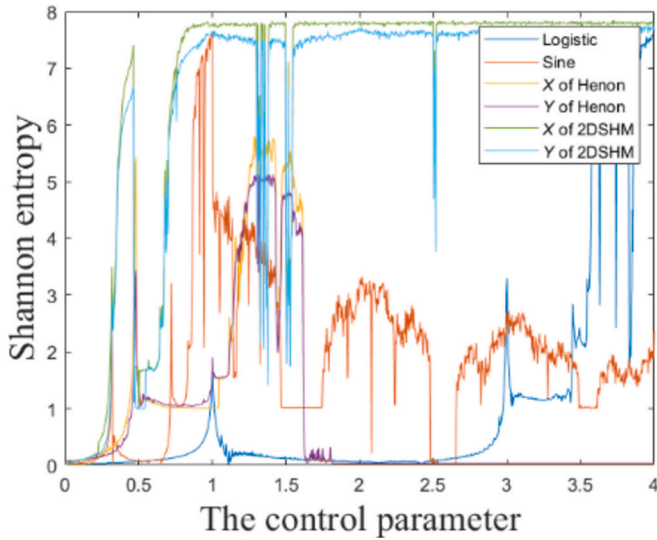


Fig. 4. Comparisons of the SEs for four chaotic maps.

2DSHM has better ergodicity, larger parameter space, and more complex chaotic behavior.

- (2) A 3D coordinate matrix is designed based on the chaotic map. Using the 3D coordinate matrix to fully scramble the multiple-image matrix can effectively eliminate the pixel correlation and improve the algorithm security.
- (3) A 3D matrix diffusion is designed based on the chaotic map. The current encrypted pixel value is affected by the pixels in its previous row, column and layer.
- (4) Using the 2DSHM and 3D coordinate matrix, a multiple-image encryption algorithm is designed, which has better encryption computational efficiency and security.

The remainder of the paper is organized as follows. In Section 2, the related works about the 2DSHM and the 3D coordinate matrix are reviewed. In Section 3, the security analysis of chaotic maps is

presented. The multiple-image encryption algorithm is designed in Section 4. In Section 5, experiments are carried out. Section 6 concludes the whole work in this paper.

2. Related works

In this section, the 2DSHM is designed and its chaotic dynamics characteristics are analyzed. Meanwhile, the 3D coordinate matrix is introduced in detail.

2.1. Classical chaotic maps

SiD chaos. It is defined as Eq. (1) [16]:

$$x_{k+1} = r \sin(\pi x_k), \quad (1)$$

where $r \in (0, 1)$ is the control parameter and $x_0 \in (0, 1)$ is the initial value.

Henon map is a simple 2D chaotic map. It is defined as Eq. (2) [17]:

$$\begin{cases} x_{k+1} = 1 + y_k - ax_k^2, \\ y_{k+1} = bx_k \end{cases}, \quad (2)$$

where $a = 1.4$ and $b = 0.3$ [40].

2.2. Definition of 2DSHM

The Sine map and Henon map have simple structures and are easy to iterate. However, they have a few parameters and initial values, low sensitivity of initial values, narrow parameter range, etc. For the desirable chaos performance, the 2DSHM is constructed. The 2DSHM couples the Sine map into the Henon map, and the modulo 1 operation is performed on the chaotic map to ensure that the chaotic value is always between 0 and 1. It is defined as Eq. (3):

$$\begin{cases} x_{k+1} = \text{mod}(1 + y_k - \mu \sin^2(\pi x_k), 1), \\ y_{k+1} = \text{mod}(\mu \sin(\pi x_k), 1) \end{cases}, \quad (3)$$

where μ is the control parameter and $\text{mod}(\cdot)$ is the modulo operation. To ensure that the 2DSHM is always in a chaotic state, $\mu \in (1, 4)$ is used in this paper.

2.3. Chaos performance analysis

The dynamical characteristics of the 2DSHM are tested and the comparisons with Sine, Logistic and Henon maps are made. The length of used chaotic sequences in all the following tests is 1000.

2.3.1. Bifurcation diagram

The bifurcation diagram can visually judge the behavior of chaotic maps. The values of the pseudo-random sequence generated by the iteration of the chaotic map should be evenly distributed within a certain range. Fig. 1 shows the bifurcation diagram of the 2DSHM. Compared with the Sine and Henon maps, the 2DSHM has stronger ergodicity.

2.3.2. Sensitivity analysis

When the parameters of a chaotic map are fixed and the initial values vary slightly, there should be some differences between the different pseudo-random sequences generated by a good chaotic map. The trajectory diagrams verify the sensitivity to the initial values for the 2DSHM and Henon maps are shown in Fig. 2. While the initial values

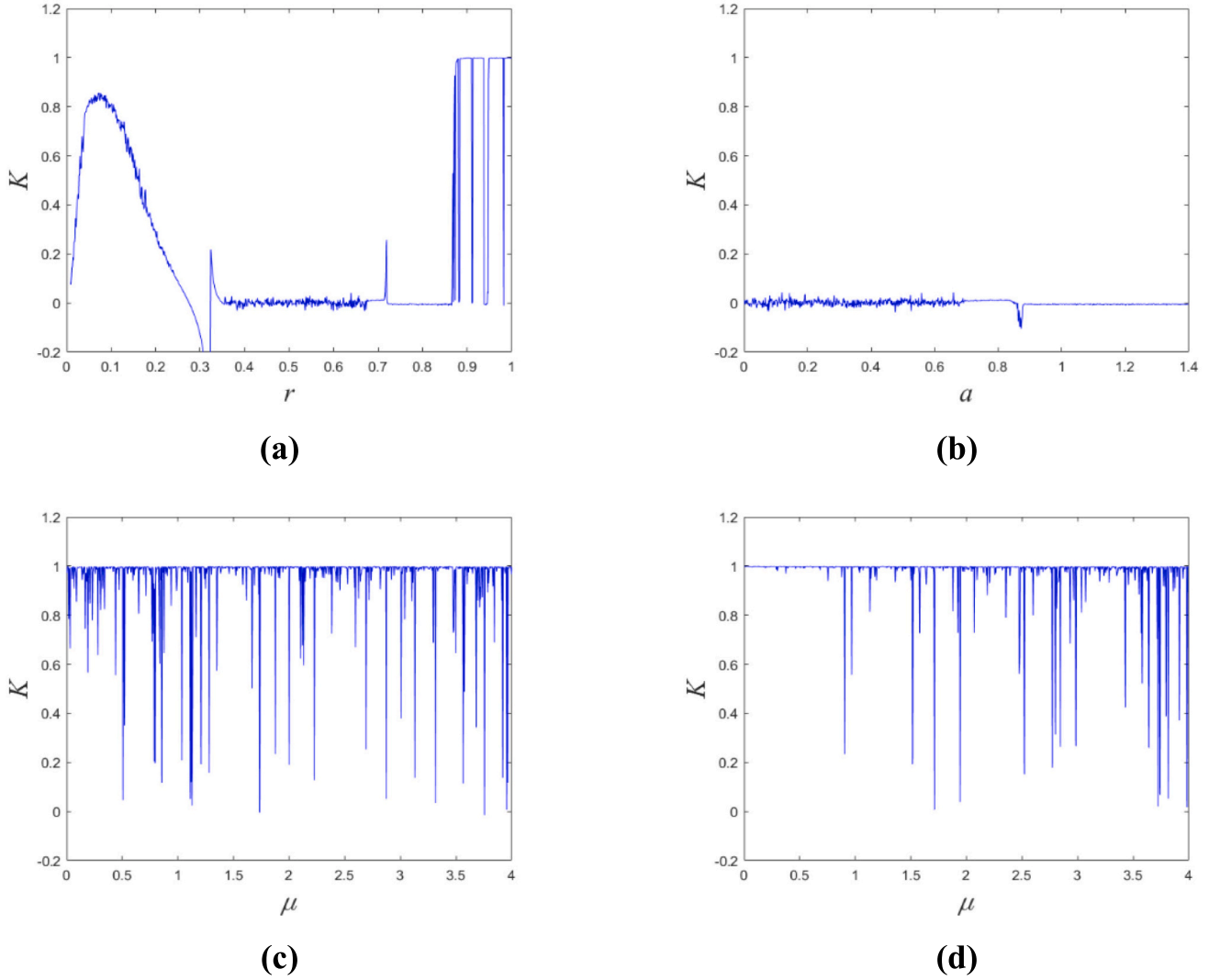


Fig. 5. 0–1 test results. (a) Sine map, (b) output X of Henon map with $b = 0.3$, (c) output X of the 2DSHM, (d) output Y of the 2DSHM.

represent by the black line are $X_1(1) = 0.7$, $Y_1(1) = 0.3$ and the initial values represent by the red line are $X_2(1) = 0.7 + 10^{-10}$, $Y_2(1) = 0.3$. Moreover, the first values of the lines shown in Fig. 2 are the initial values and are marked in blue. It can be seen that the pseudo-random sequence obtained by changing only a little bit of the initial value of the 2DSHM is completely different. According to Fig. 2, the values of the 2DSHM begin to differ significantly around $k = 15$, while the values of the Sine and Henon maps begin to differ significantly around $k = 45$. Compared with the Sine and Henon maps, the 2DSHM has the stronger sensitivity to the initial value.

2.3.3. LE

The Lyapunov exponent (LE) is commonly used to describe chaotic maps and it is defined as Eq. (4) [18]. For 1D chaotic maps, they are in a chaotic state only when the value of LE is >0 . When the value of LE is <0 , the chaotic map is in a periodic state. The MD chaotic maps have the multiple LEs. When one of the LEs is >0 , the chaotic map is in a chaotic state. What is more, the MD chaotic maps are in a hyperchaotic state if the chaotic maps have more than two LEs >0 . Fig. 3 shows the LEs of the 2DSHM, Sine, Logistic and Henon maps. The LEs of the chaotic maps will change as their control parameters change. X and Y of Henon represent two chaotic sequences generated by Henon map. Similarly, X and Y of

2DSHM represent two chaotic sequences generated by the 2DSHM. Compared with the LEs of Sine, Logistic and Henon maps, one LE of the 2DSHM is always >0 and close to 30, but the LEs of other chaotic maps are always close to 0. That means the 2DSHM is in a chaotic state and the LEs of the 2DSHM are much higher than other chaotic maps.

$$LE = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)|. \quad (4)$$

2.3.4. SE

Shannon entropy (SE) can describe the degree of chaos of pseudo-random sequences, and it is defined as Eq. (5). In this paper, the chaotic sequence is divided into 256 intervals of equal length according to the value range of the chaotic map. Then the occurrence number of the iterative value in each interval is counted, and the probability of the iterative occurrence in each interval is calculated. Finally, the SE of the sequence can be calculated. If the appearance probability of each interval appearing is equal, the SE will reach its maximum value. The theoretical maximum value of entropy is $\log_2(256) = 8$ [21]. Fig. 4 shows the 2DSHM SE and The SEs of the chaotic maps will change as their control parameters change. The green line represents the X sequence of the 2DSHM, and the blue line represents the Y sequence of

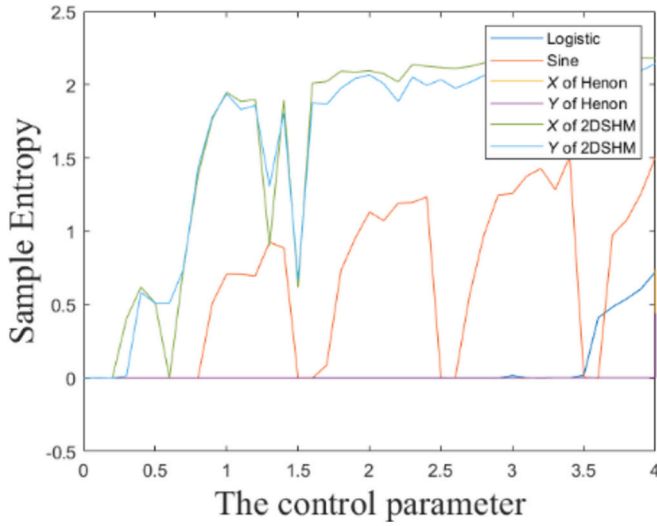


Fig. 6. Comparisons of the sample entropy for four chaotic maps.

Table 1
NIST test results.

Statistical tests	p_1	p_2	p_3	p_4	p_5	Result of the 2DSHM
Frequency	0.0000	0.1197	0.0000	0.0218	0.0356	Passed
Block frequency	0.9981	0.0077	0.0000	0.9626	0.4061	Passed
Runs	0.0000	0.0000	0.0380	0.9023	0.2231	Passed
Longest run	0.0000	0.0003	0.0035	0.2484	0.5822	Passed
Rank	0.0057	0.1143	0.0466	0.1997	0.0325	Passed
FFT	0.3840	0.8618	0.6634	0.2701	0.1105	Passed
Non-overlapping template	0.0000	0.2728	0.0000	0.6612	0.9372	Passed
Overlapping template	0.0000	0.0007	0.0000	0.4952	0.9066	Passed
Universal	0.0000	0.0014	0.3773	0.5363	0.8675	Passed
Linear complexity	0.5328	0.7498	0.2524	0.5891	0.2069	Passed
Serial test P-value 1	0.0000	0.0000	0.0000	0.0126	0.0457	Passed
Serial test P-value 2	0.0000	0.0000	0.0000	0.0564	0.0228	Passed
Approximate entropy	0.0276	0.0558	0.0000	0.7741	0.0325	Passed
Cumulative sums-forward	0.9992	0.8588	0.0005	0.7505	0.0328	Passed
Cumulative sums-reverse	1.0000	0.8973	0.0011	0.9999	0.0302	Passed
Random excursions test ($X = 1$)	0.8131	0.2519	0.0103	0.7082	0.2178	Passed
Random excursions variant test ($X = 1$)	0.9389	0.4601	0.2299	0.2877	0.6255	Passed

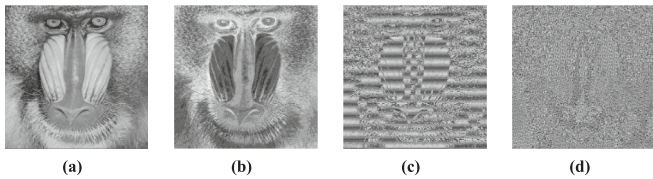


Fig. 7. XOR results of different chaotic maps. (a) Plain image, (b) XOR by the Henon map, (c) XOR by the Sine map, (d) XOR by the 2DSHM.

the 2DSHM. Compared with the Sine, Logistic and Henon chaotic maps, the SEs of the 2DSHM are closer to 8 in the parameter range of $[0, 4]$ and the SEs of other chaotic maps are no >6 , so the 2DSHM has better randomness.

$$SE = - \sum_{i=0}^n p(s_i) \log_2 p(s_i). \quad (5)$$

2.3.5. 0–1 test

The 0–1 test can determine the existence of chaos by calculating the linear growth rate $K(c)$ value of the variable after discrete data transformation. For a discrete time series $\theta(n)$, $n = 1, 2, \dots, N$ and any random number $c \in (0, \pi)$, the test value $K(c)$ is calculated as follows:

$$K(c) = \frac{\text{corr}(\xi, \Delta)}{\sqrt{\text{var}(\xi)\text{var}(\Delta)}} \in [-1, 1], \quad (6)$$

where

$$\xi = 1, 2, \dots, n, \Delta = D_c(1), D_c(2), \dots, D_c(n),$$

$$\text{cov}(x, y) = \frac{1}{q} \sum_{i=1}^q (x(i) - \bar{x})(y(i) - \bar{y}),$$

$$\bar{x} = \frac{1}{q} \sum_{i=1}^q x(i),$$

$$\text{var}(x) = \text{cov}(x, x),$$

$$D_c(n) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \left((p_c(i+n) - p_c(i))^2 + (q_c(i+n) - q_c(i))^2 \right) - (E(\theta))^2 \frac{1 - \cos(nc)}{1 - \cos(c)},$$

and

$$p_c(n) = \sum_{i=1}^n \theta(i) \cos(ic),$$

$$q_c(n) = \sum_{i=1}^n \theta(i) \sin(ic),$$

$$E(\theta) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \theta(i).$$

The closer the value of $K(c)$ is to 1, the sequence is chaotic. Let $N = 100$ and $c \in (\pi/5, 4\pi/5)$. Fig. 5 illustrates the results of the 0–1 test. $K(c)$ values of Sine map are close to 1 within a small range and $K(c)$ values of Henon map are close to 0. However, $K(c)$ values of the 2DSHM are close to 1 most of the time. Thus, the 2DSHM has better chaotic properties.

2.3.6. Sample entropy

Sample entropy can be used to measure the time complexity of a sequence. The smaller the value of sample entropy, the higher the self-similarity of the sequence; the larger the value of sample entropy, the more complex the sample sequence and the better the randomness. Fig. 6 shows the sample entropy of the 2DSHM, Sine, Logistic and Henon maps. The Sample entropy of the chaotic maps will change as their control parameters change. Regardless of the value of the parameter, the sample entropy of the 2DSHM is always higher than that of the other two chaotic maps. Therefore, the 2DSHM has higher time complexity.

2.3.7. NIST test

National Institute of Standards and Technology (NIST) test evaluates the random number generator, and then uses the results to assess whether the chaotic system is suitable for the encryption scheme. The NIST test contains 15 tests, and each test method generates a P -value. When the P -value is >0.01 , the sequence can be considered random. Multiple sequences are generated by the 2DSHM and then convert these sequences into binary bit streams for testing. Table 1 shows the test results of Sine, Henon map and the 2DSHM. p_1 is P -values of Sine map, p_2

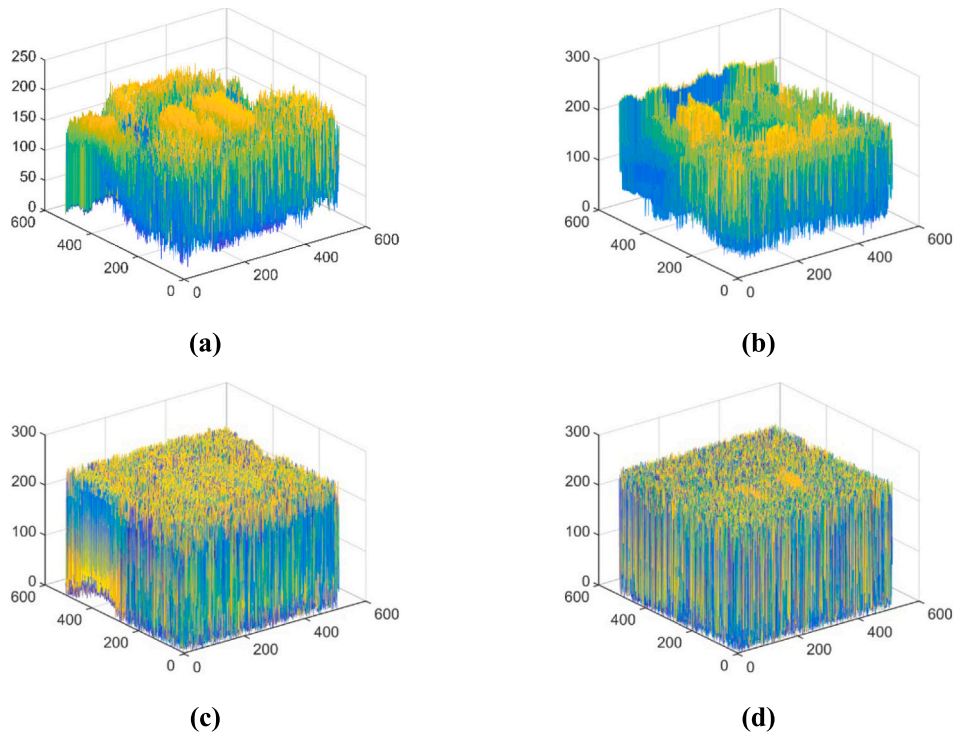


Fig. 8. Histogram analysis of different chaotic maps. (a) Plain image, (b) XOR by the Henon map, (c) XOR by the Sine map, (d) XOR by the 2DSHM.

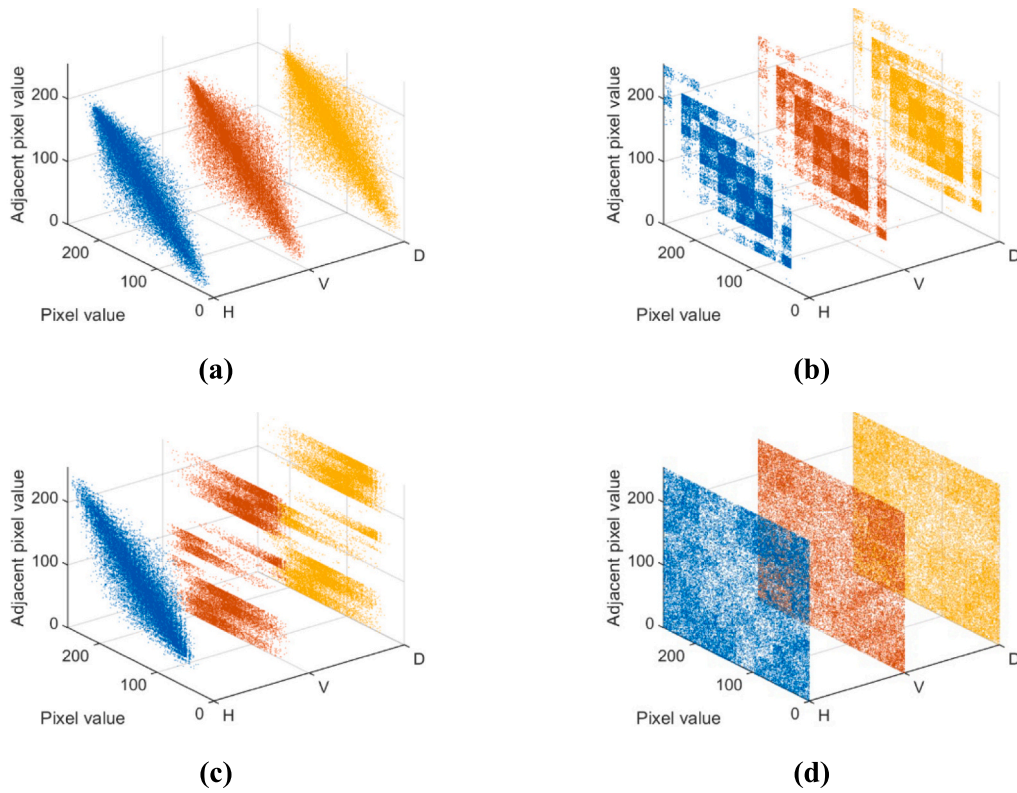


Fig. 9. Pixel correlation of different chaotic maps. (a) Plain image, (b) XOR by the Henon map, (c) XOR by the Sine map, (d) XOR by the 2DSHM.

Table 2
Correlation coefficients for different chaotic maps.

Chaotic map	Correlation coefficients		
	Horizontal	Vertical	Diagonal
Henon	0.7509	0.6511	0.6153
Sine	0.6606	-0.3907	-0.3696
2DSHM	-0.0464	0.0075	0.0719

and p_3 are P -values of Henon map. However, there are some P -values that are not >0.001 , which means they do not pass some tests. p_4 and p_5 are P -values of the 2DSHM, and each bitstream of them passes all the tests. It indicates that the pseudo-random sequences generated by the 2DSHM are highly random.

3. Chaotic map security analysis

In this section, we analyze and compare the impact of the 2DSHM, Sine and Henon maps on the security of image encryption. The security of image encryption algorithms is related to many factors, such as chaos performance, specific scrambling scheme, specific diffusion scheme, and the number of encryption rounds. The exclusive OR (XOR) is a very common encryption operation. To well reflect the impact of chaotic maps on the security of image encryption algorithms, only the XOR operation is used for encrypting the test image in our experiments, i.e., using chaotic sequence to perform the XOR operation on image pixels. The initial values and control parameters of the three chaotic maps in the security analyses are the same and the test image is baboon. The initial values of the 2DSHM and the Henon map are $x_0 = 0.7$ and $y_0 = 0.3$, respectively, and the initial value of the Sine map is $x_0 = 0.7$. The values of the control parameters are $r = 0.8$, $a = 0.8$, $b = 0.3$ and $\mu = 0.8$. Fig. 7 shows the plain image and encrypted images generated by the three chaotic maps. From these images, we can see that Fig. 7(d) has the best encryption effect.

3.1. Histogram analysis

The change in lightness and darkness of the image can be visualized from the histogram of the image and it is a common method for statistical analysis of images. The pixel value distribution of an excellent encrypted image should be average. Fig. 8 shows the histogram of the plain image and encrypted images. Compared with other chaotic maps, the pixel value distribution in Fig. 8(d) is more even.

3.2. Correlation analysis

The correlation analysis refers to the analysis of two or more correlated variable elements. The pixels of the plain images exist a strong correlation between neighboring pixels. Therefore, a secure image encryption algorithm should reduce the correlation between adjacent pixels as much as possible and bring the correlation coefficient close to 0. Fig. 9 shows the pixel correlation comparative analyses. It can be seen that the 2DSHM makes the image pixels almost have no correlation in all directions, while the image pixels encrypted by the other two chaotic maps have high correlation. Meanwhile, we have presented the correlation coefficients in Table 2, which show the superiority of the 2DSHM.

4. Proposed multiple-image encryption algorithm

In this section, a new multiple-image encryption algorithm using the newly constructed chaos is proposed. The encryption algorithm is divided into three stages, i.e., keys generation, scrambling and diffusion. The flowchart of the image encryption process is shown in Fig. 10.

4.1. Key generation

In this stage, the SHA-256 algorithm is first utilized to produce the hash values of plain image. And then, we can calculate the initial conditions of the 2DSHM using these hash values. Finally, two chaotic sequences X , Y are produced by the 2DSHM. Algorithm 1 describes the key generation process.

Step 1: Hash value generation

The image sizes of the plain image set $P = \{P_1, P_2, \dots, P_k\}$ are $\{m_1 \times n_1, m_2 \times n_2, \dots, m_k \times n_k\}$ respectively and $m = \text{MAX}\{m_1, m_2, \dots, m_k\}$, $n = \text{MAX}\{n_1, n_2, \dots, n_k\}$. The SHA-256 algorithm is first utilized to generate a 256-bit digest H of a random plain image from the set P . After that, we convert each group of 8 bits H in the hash value into a decimal number. Finally, we can get 32 hash values in decimal form, recorded as $H = (h_1, h_2, \dots, h_{32})$.

Step 2: Key generation

The initial conditions of the 2DSHM are calculated from these hash values $h_1, h_2, \dots, h_{32}$. The specific calculation is shown in Eq. (7).

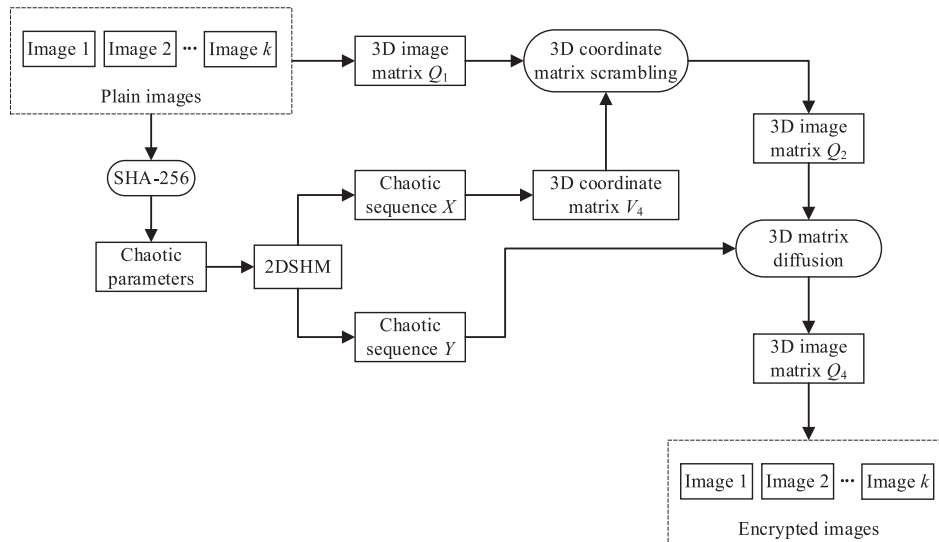


Fig. 10. Flow chart of image encryption process.

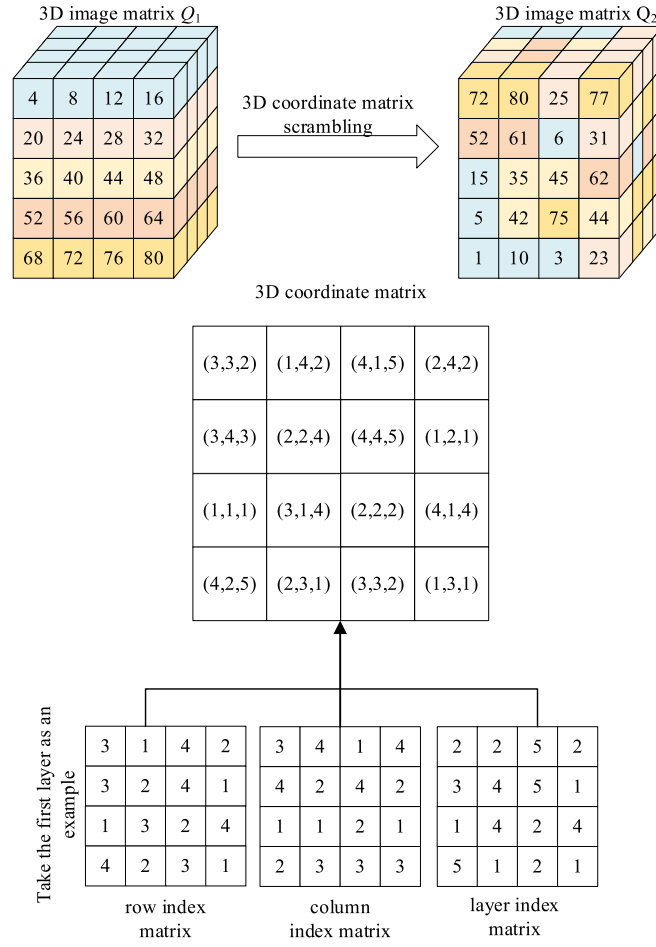


Fig. 11. 3D coordinate matrix scrambling.

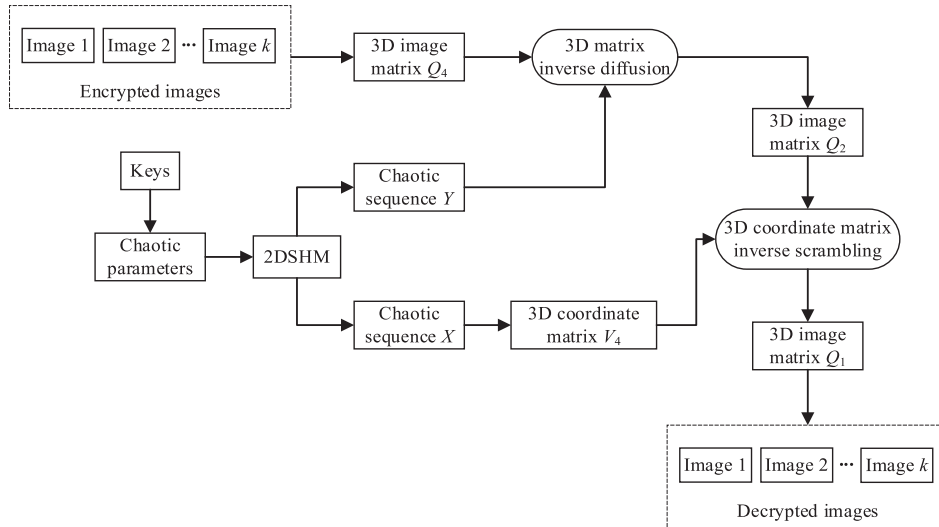


Fig. 12. Flow chart of image decryption process.

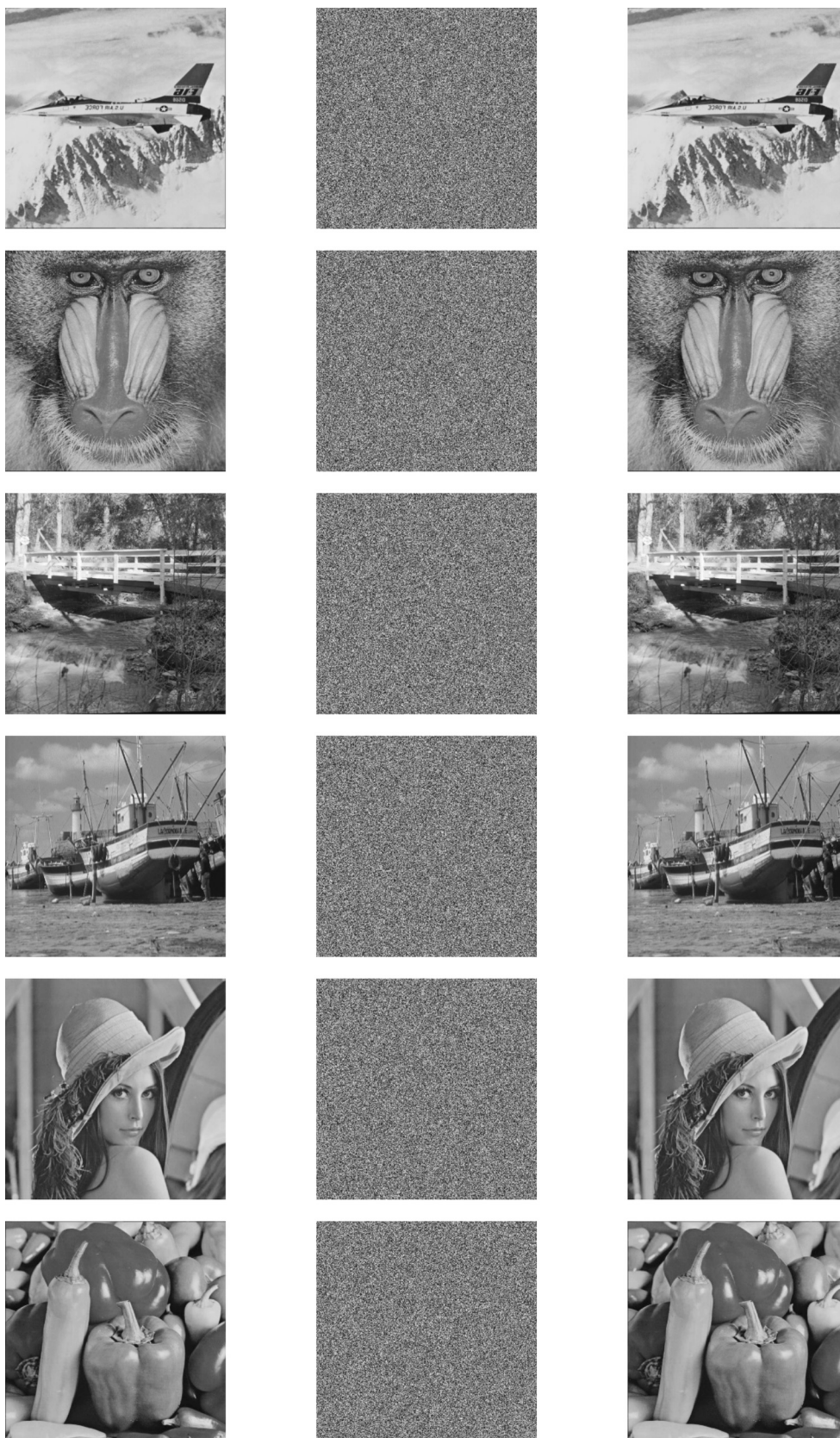


Fig. 13. Examples of image encryption and decryption. (left column): plain images, (center column): encrypted images, (right column): decrypted images (top to down: Airplane, Baboon, Bridge, Fishing boat, Lena and Peppers).

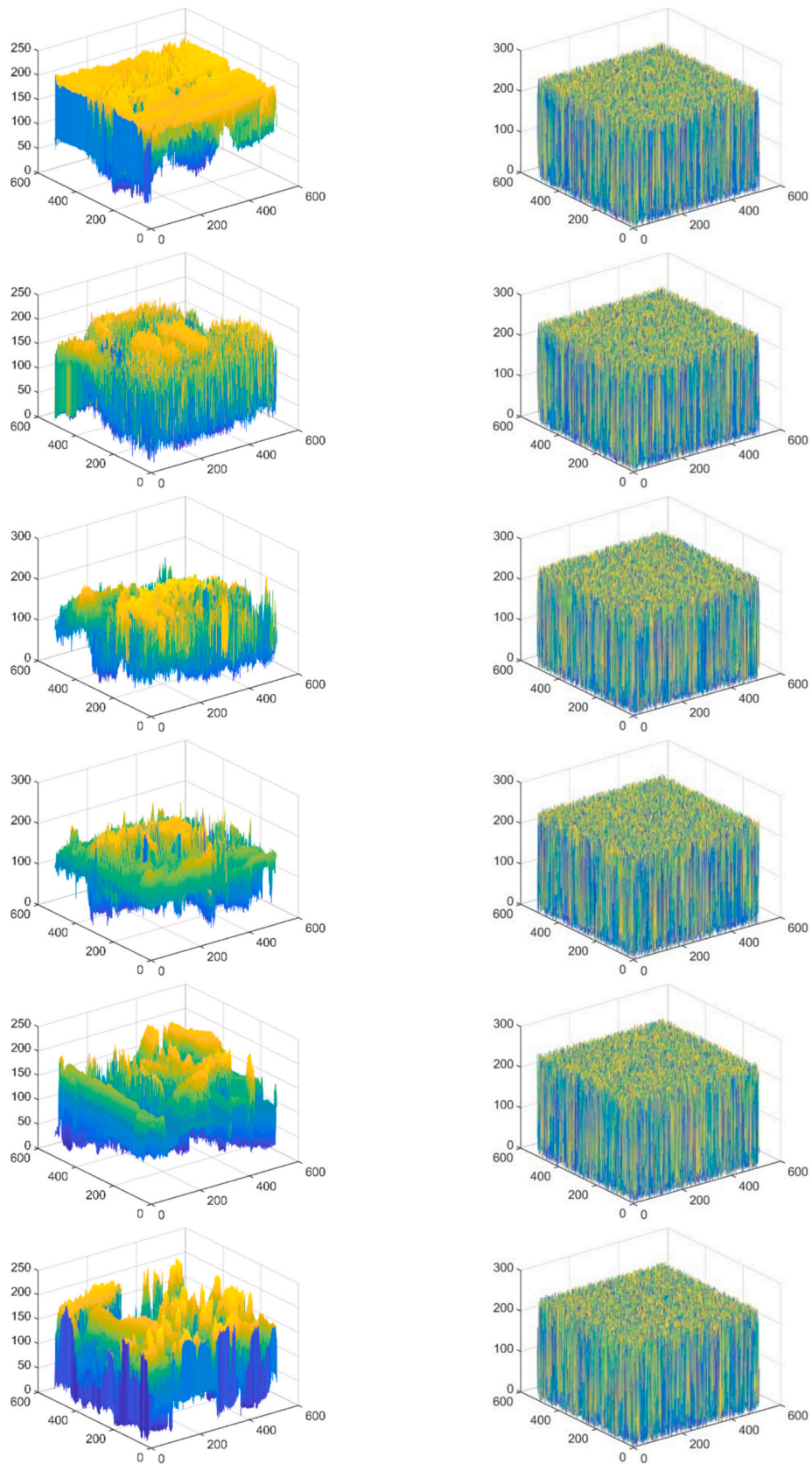


Fig. 14. Histogram analysis (left column): plain images, (right column): encrypted images (top to down: Airplane, Baboon, Bridge, Fishing boat, Lena and Peppers).

$$\begin{cases} \mu = \frac{h_1 \oplus h_5 \oplus h_9 \oplus h_{13}}{256} + \frac{h_2 \oplus h_6 \oplus h_{10} \oplus h_{14}}{256} + \frac{h_3 \oplus h_7 \oplus h_{11} \oplus h_{15}}{256} \\ x_0 = \frac{h_4 \oplus h_8 \oplus h_{17} \oplus h_{19} \oplus h_{21} \oplus h_{23} \oplus h_{25} \oplus h_{27} \oplus h_{29} \oplus h_{31}}{256} \\ y_0 = \frac{h_{12} \oplus h_{16} \oplus h_{18} \oplus h_{20} \oplus h_{22} \oplus h_{24} \oplus h_{26} \oplus h_{28} \oplus h_{30} \oplus h_{32}}{256} \end{cases}, \quad (7)$$

where $\oplus$ is the XOR operation.

Step 3: Chaotic sequence generation

The 2DSHM is iterated $1000 + k \times m \times n$ times based on the initial values μ, x_0, y_0 and the first 1000 values of the 2DSHM are discarded. Two chaotic sequences X and Y are generated with the same length of $k \times m \times n$.

Algorithm 1. Key generation process.

Input: the plain image set P and the max size $m \times n$ of plain image set P
Output: the chaotic sequences X and Y

1: /* Get the hash value of the plain image and group it */
2: $K \leftarrow \text{SHA256}(P)$
3: $\{k_1, k_2, \dots, k_{32}\} \leftarrow K$
4: $\{h_1, h_2, \dots, h_{32}\} \leftarrow \text{bin2dec}\{k_1, k_2, \dots, k_{32}\}$
5: /* Calculate the control parameters and initial values μ, x_0 and y_0 */
6: $\mu \leftarrow 1 + (((h_1 + h_5 + h_9 + h_{13}) \oplus (h_2 + h_6 + h_{10} + h_{14}) \oplus (h_3 + h_7 + h_{11} + h_{15})) / 1024)$
7: $x_0 \leftarrow (h_4 \oplus h_8 \oplus h_{17} \oplus h_{19} \oplus h_{21} \oplus h_{23} \oplus h_{25} \oplus h_{27} \oplus h_{29} \oplus h_{31}) / 256$
8: $y_0 \leftarrow (h_{12} \oplus h_{16} \oplus h_{18} \oplus h_{20} \oplus h_{22} \oplus h_{24} \oplus h_{26} \oplus h_{28} \oplus h_{30} \oplus h_{32}) / 256$
9: /* Calculate the chaotic sequences X and Y */
10: X and $Y \leftarrow \text{2DSHM}(\mu, x_0, y_0)$

Remark: Notations are represented by '/* */'.

4.2. 3D coordinate matrix scrambling

In this stage, the 3D coordinate matrices are generated by the chaotic sequence X . These matrices are used to scramble the plain images of the set P . The specific scrambling process of 3D coordinate matrix scrambling is shown in Fig. 11.

Step 1: Plain images preprocessing

k images of P are enlarged into $m \times n$. The expanded pixel values are all "0". For example, three images with the different sizes, 480×480 , size 500×500 and 512×512 are encrypted at the same time. Therefore, $m = 512, n = 512$. For the 480×480 image, 32 columns are added to the right side of this image and 32 rows are added to the bottom of this image. All the added pixel values are "0". Q_1 is a 3D matrix of the size $k \times m \times n$ generated by reshaping P and each layer of Q_1 is a plain image.

Step 2: Chaotic sequence preprocessing

The 3D matrix V of the size $k \times m \times n$ is obtained by reshaping X . After that the rows, columns, and layers of each matrix of V is indexed to obtain three 3D matrices V_1, V_2 and V_3 .

Step 3: 3D matrices generation

The pixel values at the same positions of V_1, V_2 and V_3 are combined into one coordinate to obtain the 3D coordinate matrix V_4 .

Step 4: 3D coordinate matrix scrambling

By the 3D coordinate matrix V_4 , the elements in Q_1 are sequentially exchanged and the scrambled 3D matrix is defined as Q_2 . The specific process is shown in Eq. (8).

$$\begin{cases} a = Q_2(i, j, t) \\ Q_2(i, j, t) = Q_2(V_4(i, j, t)) \\ Q_2(V_4(i, j, t)) = a \end{cases} \quad (8)$$

Taking the $4 \times 4 \times 5$ image matrix as an example, the row, column and layer index matrices are generated from the 3D chaos matrix. The

three index matrices are combined to generate a 3D coordinate matrix. Then starting from $Q_1(1, 1, 1)$, the pixels in Q_1 swap positions with the pixels of corresponding coordinates in the 3D coordinate matrix and the scrambled 3D image matrix Q_2 is generated.

4.3. 3D matrix diffusion

3D matrix scrambling operation can only change the pixel position, but it cannot change the pixel value. To achieve the excellent encryption effect, the proposed multiple-image encryption algorithm will use the 3D matrix diffusion operation to change the pixel value. Algorithm 2 describes the specific encryption process.

Algorithm 2. Encryption process.

Input: the plain image set P , the max size $m \times n$ of plain image set P , the number of the plain image k and the chaotic sequences X and Y generated in the Algorithm 1
Output: the encrypted image set C

1: /* 3D coordinate matrix scrambling */
2: Expand the size of all images in P to $m \times n$
3: $Q_1 \leftarrow \text{reshape}(P, m, n, k)$
4: $V \leftarrow \text{reshape}(X, m, n, k)$
5: Index the rows, columns, and layers of each matrix of V to obtain three 3D matrices V_1, V_2 and V_3
6: combine V_1, V_2 and V_3 to obtain V_4
7: $Q_2 \leftarrow Q_1$
8: for $i \leftarrow 1$ to m do
9: for $j \leftarrow 1$ to n do
10: for $t \leftarrow 1$ to k do
11: $a \leftarrow Q_2(i, j, t)$
12: $Q_2(i, j, t) \leftarrow Q_2(V_4(i, j, t))$
13: $Q_2(V_4(i, j, t)) \leftarrow a$
14: end
15: end
16: end
17: /* 3D matrix diffusion */
18: $S_1 \leftarrow \text{mod}(\text{floor}(Y \times 10^{10}), 256)$
19: $S_2 \leftarrow \text{mod}(\text{ceil}(Y \times 10^{12}), 256)$
20: $S_1 \leftarrow \text{reshape}(S_1, m, n, k)$
21: $S_2 \leftarrow \text{reshape}(S_2, m, n, k)$
22: for $i \leftarrow 1$ to m do
23: for $j \leftarrow 1$ to n do
24: for $t \leftarrow 1$ to k do
25: if $i=1$ and $j=1$ then
26: $Q_3(i, j, t) \leftarrow Q_2(i, j, t) \oplus S_1(i, j, t)$
27: else
28: if $i=1$ and $j \neq 1$ then
29: $Q_3(i, j, t) \leftarrow Q_2(i, j, t) \oplus Q_3(i, j-1, t) \oplus S_1(i, j, t)$
30: else
31: if $j=1$ and $i \neq 1$ then
32: $Q_3(i, j, t) \leftarrow Q_2(i, j, t) \oplus Q_3(i-1, j, t) \oplus S_1(i, j, t)$
33: else
34: if $i \neq 1, j \neq 1$ and $j \neq 1$ then
35: $Q_3(i, j, t) \leftarrow Q_2(i, j, t) \oplus Q_3(i-1, j-1, t) \oplus S_1(i, j, t)$
36: else
37: $Q_3(i, j, t) \leftarrow Q_2(i, j, t) \oplus Q_3(i-1, j-1, t) \oplus Q_3(i, j, t-1) \oplus S_1(i, j, t)$
38: end
39: end
40: end
41: Similarly, using Eq. (11) to get Q_4
42: for $i \leftarrow 1$ to k do
43: $C(i) \leftarrow Q_4(:, :, i)$
44: end

Step 1: Chaotic sequence preprocessing

By applying Eqs. (9) and (10) to the chaotic sequence Y , the chaotic sequences S_1 and S_2 are obtained. After that, we reshape the S_1 and S_2 into 3D matrices of the size $k \times m \times n$.

$$S_1 = \text{mod}(\text{floor}(Y \times 10^{10}), 256), \quad (9)$$

$$S_2 = \text{mod}(\text{ceil}(Y \times 10^{12}), 256), \quad (10)$$

where $\text{floor}(\cdot)$ is the downward rounding function and $\text{ceil}(\cdot)$ is the rounding up operation.

Step 2: Forward 3D matrix diffusion

The 3D image matrix Q_3 is generated by performing the XOR operation for image diffusion. The detailed diffusion process is shown in Eq. (11). The diffusion operation starts from $Q_2(1, 1, 1)$ and ends at $Q_2(m, n, k)$.

Step 3: Reverse 3D matrix diffusion

The 3D image matrix Q_4 is generated by performing the XOR operation for image diffusion. The detailed diffusion process is shown in Eq. (12). The diffusion operation starts from $Q_3(m, n, k)$ and ends at $Q_3(1, 1, 1)$.

Step 4: Final encrypted images

Treat each layer of Q_4 as an encrypted image to obtain the final encrypted image set $C = \{C_1, C_2, \dots, C_k\}$.

$$\begin{cases} Q_3(i,j,t) = Q_2(i,j,t) \oplus S_1(i,j,t), & i=1, j=1, t \in (1,k) \\ Q_3(i,j,t) = Q_2(i,j,t) \oplus Q_3(i,j-1,t) \oplus S_1(i,j,t), & i=1, j \in (2,n), t \in (1,k) \\ Q_3(i,j,t) = Q_2(i,j,t) \oplus Q_3(i-1,j,t) \oplus S_1(i,j,t), & j=1, i \in (2,m), t \in (1,k) \\ Q_3(i,j,t) = Q_2(i,j,t) \oplus Q_3(i-1,j-1,t) \oplus S_1(i,j,t), & t=1, i \in (2,m), j \in (2,n) \\ Q_3(i,j,t) = Q_2(i,j,t) \oplus Q_3(i-1,j-1,t) \oplus Q_3(i,j,t-1) \oplus S_1(i,j,t), & \text{others} \end{cases} \quad (11)$$

$$\begin{cases} Q_4(i,j,t) = Q_3(i,j,t) \oplus S_2(i,j,t), & i=m, j=n, t \in (1,k) \\ Q_4(i,j,t) = Q_3(i,j,t) \oplus Q_4(i,j+1,t) \oplus S_2(i,j,t), & i=m, j \in (1,n-1), t \in (1,k) \\ Q_4(i,j,t) = Q_3(i,j,t) \oplus Q_4(i+1,j,t) \oplus S_2(i,j,t), & j=n, i \in (1,m-1), t \in (1,k) \\ Q_4(i,j,t) = Q_3(i,j,t) \oplus Q_4(i+1,j+1,t) \oplus S_2(i,j,t), & t=k, i \in (1,m-1), j \in (1,n-1) \\ Q_4(i,j,t) = Q_3(i,j,t) \oplus Q_4(i+1,j+1,t) \oplus Q_3(i,j,t+1) \oplus S_2(i,j,t), & \text{others} \end{cases} \quad (12)$$

4.4. Decryption algorithm

The decryption algorithm is the reverse process of the encryption algorithm. The sender transmits the encrypted images and keys to the recipient, and the decryption algorithm generates chaotic sequences X and Y through the keys. The decrypted images are obtained by performing the inverse 3D matrix diffusion and inverse 3D coordinate matrix scrambling operations on the encrypted images using the chaotic sequences. The flowchart of the decryption algorithm is shown in Fig. 12, and Algorithm 3 describes the specific decryption process.

Step 1: Chaotic sequence generation

The 2DSHM is iterated $1000 + k \times m \times n$ times based on the initial values μ, x_0, y_0 and the first 1000 values of 2DSHM are discarded. Two chaotic sequences X and Y are generated with the same length of $k \times m \times n$.

Step 2: 3D matrices generation

The 3D matrix V is obtained by reshaping X into a 3D matrix of the size $k \times m \times n$. After that we index the rows, columns, and layers of each matrix of V to obtain three 3D matrices V_1, V_2 and V_3 . Meanwhile, the chaotic sequences S_1 and S_2 is obtained by Eqs. (9) and (10). After that, we reshape the S_1 and S_2 into a 3D matrix of the size $k \times m \times n$.

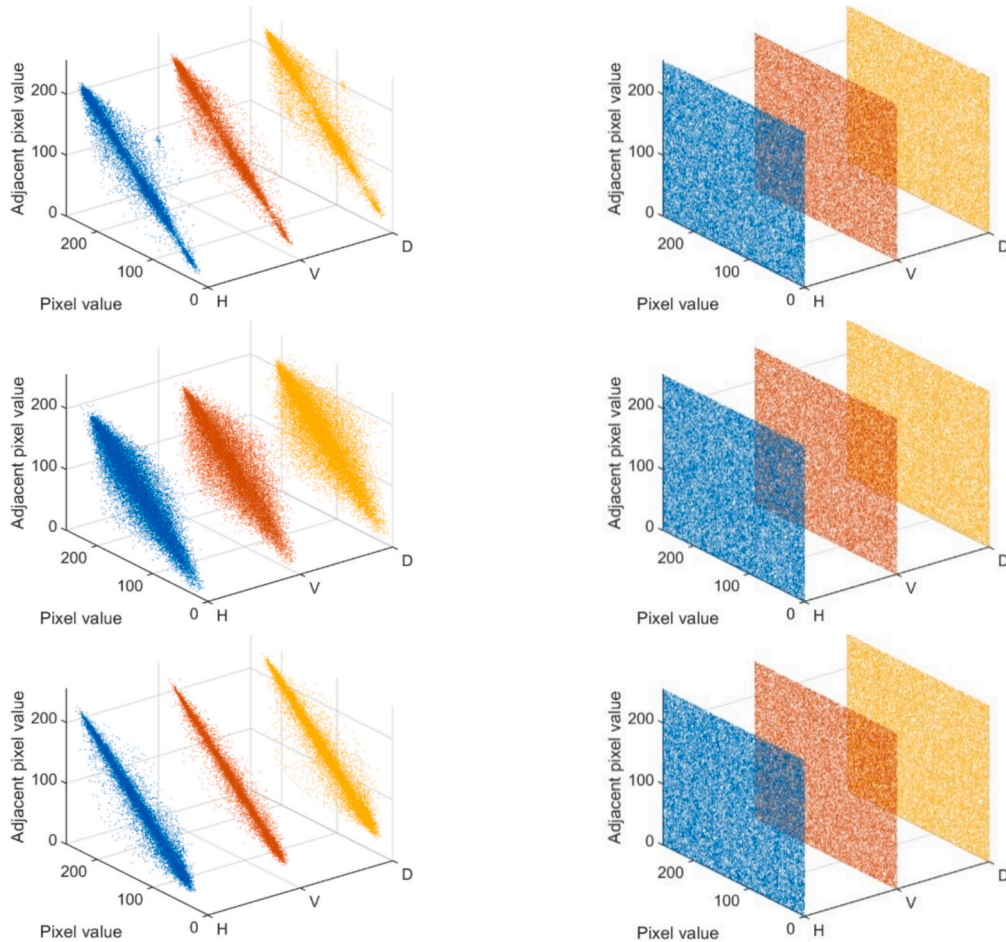


Fig. 15. Pixel correlation between the plain images and their encrypted images in various direction (left column): plain images, (right column): encrypted images (top to down: Airplane, Baboon, Lena).

Step 3: Encrypted images preprocessing

Q_4 is generated by reshaping C as a 3D matrix of the size $k \times m \times n$, and each layer of Q_4 is an encrypted image.

Step 4: Inverse 3D matrix diffusion

The 3D image matrix Q_2 is generated by preforming the XOR operation for image diffusion. The detailed diffusion is shown in Eqs. (13) and (14).

$$\begin{cases} Q_3(i, j, t) = Q_4(i, j, t) \oplus S_2(i, j, t), & i = m, j = n, t \in (1, k) \\ Q_3(i, j, t) = Q_4(i, j, t) \oplus Q_4(i, j+1, t) \oplus S_2(i, j, t), & i = m, j \in (1, n-1), t \in (1, k) \\ Q_3(i, j, t) = Q_4(i, j, t) \oplus Q_4(i+1, j, t) \oplus S_2(i, j, t), & j = n, i \in (1, m-1), t \in (1, k) \\ Q_3(i, j, t) = Q_4(i, j, t) \oplus Q_4(i+1, j+1, t) \oplus S_2(i, j, t), & t = k, i \in (1, m-1), j \in (1, n-1) \\ Q_3(i, j, t) = Q_4(i, j, t) \oplus Q_4(i+1, j+1, t) \oplus Q_4(i, j, t+1) \oplus S_2(i, j, t), & \text{others} \end{cases} \quad (13)$$

$$\begin{cases} Q_2(i, j, t) = Q_3(i, j, t) \oplus S_1(i, j, t), & i = 1, j = 1, t \in (1, k) \\ Q_2(i, j, t) = Q_3(i, j, t) \oplus Q_3(i, j-1, t) \oplus S_1(i, j, t), & i = 1, j \in (2, n), t \in (1, k) \\ Q_2(i, j, t) = Q_3(i, j, t) \oplus Q_3(i-1, j, t) \oplus S_1(i, j, t), & j = 1, i \in (2, m), t \in (1, k) \\ Q_2(i, j, t) = Q_3(i, j, t) \oplus Q_3(i-1, j-1, t) \oplus S_1(i, j, t), & t = 1, i \in (2, m), j \in (2, n) \\ Q_2(i, j, t) = Q_3(i, j, t) \oplus Q_3(i-1, j-1, t) \oplus Q_3(i, j, t-1) \oplus S_1(i, j, t), & \text{others} \end{cases} \quad (14)$$

Table 3
Correlation coefficients of different images.

Plain image	Test image	Horizontal	Vertical	Diagonal
Airplane	Plain image	0.9621	0.9592	0.9320
	Encrypted image	0.0033	0.0009	0.0031
Baboon	Plain image	0.8611	0.7451	0.7120
	Encrypted image	0.0010	-0.0002	-0.0006
Bridge	Plain image	0.9384	0.9231	0.8942
	Encrypted image	-0.0017	-0.0003	0.0011
Fishing boat	Plain image	0.9401	0.9743	0.9222
	Encrypted image	-0.0002	-0.0002	-0.0003
Lena	Plain image	0.9732	0.9864	0.9600
	Encrypted image	0.0027	-0.0023	-0.0018
Peppers	Plain image	0.9767	0.9795	0.9625
	Encrypted image	-0.0013	0.0016	0.0013
Airplane [22]	Encrypted image	-0.0089	-0.0066	0.0015
Baboon [22]	Encrypted image	0.0023	-0.0033	0.0092
Baboon [23]	Encrypted image	0.0005	0.0020	0.0058
Baboon [31]	Encrypted image	-0.0073	-0.0149	-0.0018
Peppers [31]	Encrypted image	-0.0031	-0.0131	0.0020
Lena [32]	Encrypted image	0.0026	0.0050	0.0020

Table 4
Information entropy of plain images and encrypted images.

Test image	Plain image	Encrypted image
Airplane	6.7089	7.9993
Baboon	7.3664	7.9994
Bridge	7.7237	7.9993
Fishing boat	7.2092	7.9993
Lena	7.4417	7.9994
Peppers	7.5821	7.9993
Fishing boat [23]	-	7.9992
Peppers [23]	-	7.9993
Lena [24]	-	7.9993
Peppers [24]	-	7.9993
Baboon [24]	-	7.9993
Lena [25]	-	7.9993
Peppers [25]	-	7.9993
Baboon [25]	-	7.9993
Lena [33]	-	7.9993
Peppers [33]	-	7.9993
Baboon [33]	-	7.9993
Peppers [34]	-	7.9993
Fishing boat [34]	-	7.9993

Step 5: 3D coordinate matrix generation

The pixel values at the same positions of V_1 , V_2 and V_3 are combined into one coordinate to obtain the 3D coordinate matrix V_4 .

Step 6: Inverse 3D coordinate matrix scrambling

By the 3D coordinate matrix V_4 , the elements in Q_2 are sequen-

tially exchanged and the scrambled 3D matrix is Q_1 . The specific process is shown in Eq. (15).

$$\begin{cases} a = Q_1(i, j, t) \\ Q_1(i, j, t) = Q_1(V_4(i, j, t)) \\ Q_1(V_4(i, j, t)) = a \end{cases} \quad (15)$$

Step 7: Final decrypted images generation

Treat each layer of Q_1 as a decrypted image to obtain the final decrypted image set $P = \{P_1, P_2, \dots, P_k\}$.

Algorithm 3. Decryption process.

Input: the encrypted image set C , the size $m \times n$ of encrypted image set C , number of the encrypted image k and the initial values μ, x_0, y_0	23: else
Output: the decrypted image set P	24: $Q_3(i, j, t) \leftarrow Q_4(i, j, t) \oplus$
1: /* Generate the chaotic sequences */	$Q_4(i+1, j+1, t) \oplus Q_4(i, j, t+1) \oplus S_2(i, j, t)$
2: Use Algorithm 1 to generate chaotic sequences X and Y	25: end
3: /*3D matrix inverse diffusion */	26: end
4: $Q_3 \leftarrow \text{reshape}(C, m, n, k)$	27: end
5: $S_1 \leftarrow \text{mod}(\text{floor}(Y \times 10^{10}), 256)$	28: Similarly, using Eq. (13) to get Q_2
6: $S_2 \leftarrow \text{mod}(\text{ceil}(Y \times 10^{12}), 256)$	29: /* Inverse 3D coordinate matrix scrambling */
7: $S_1 \leftarrow \text{reshape}(S_1, m, n, k)$	30: $V \leftarrow \text{reshape}(X, m, n, k)$
8: $S_2 \leftarrow \text{reshape}(S_2, m, n, k)$	31: Index the rows, columns, and layers of each matrix of V to obtain three 3D matrices V_1, V_2 and V_3
9: for $i \leftarrow 1$ to m do	32: combine V_1, V_2 and V_3 to obtain V_4
10: for $j \leftarrow 1$ to n do	33: $Q_1 \leftarrow Q_2$
11: for $t \leftarrow 1$ to k do	34: for $i \leftarrow 1$ to m do
12: if $i = m$ and $j = n$ then	35: for $j \leftarrow 1$ to n do
13: $Q_3(i, j, t) \leftarrow Q_4(i, j, t) \oplus S_2(i, j, t)$	36: for $t \leftarrow 1$ to k do
14: else	37: $a \leftarrow Q_1(m+1-i, n+1-j, k+1-t)$
15: if $i = m$ and $j \neq n$ then	38: $Q_1(m+1-i, n+1-j, k+1-t) \leftarrow Q_1(V_4(m+1-i, n+1-j, k+1-t))$
16: $Q_3(i, j, t) \leftarrow Q_4(i, j, t) \oplus Q_4(i, j+1, t) \oplus S_2(i, j, t)$	39: $Q_1(V_4(m+1-i, n+1-j, k+1-t)) \leftarrow a$
17: else	40: end
18: if $j = n$ and $i \neq m$ then	41: end
19: $Q_3(i, j, t) \leftarrow Q_4(i, j, t) \oplus Q_4(i+1, j, t) \oplus S_2(i, j, t)$	42: end
20: else	43: for $i \leftarrow 1$ to k do
21: if $t = k, j \neq n$ and $i \neq m$ then	44: $P(i) \leftarrow Q_1(:, :, i)$
22: $Q_3(i, j, t) \leftarrow Q_4(i, j, t) \oplus Q_4(i+1, j+1, t) \oplus S_2(i, j, t)$	45: end

Table 5

The results of NPCR and UACI.

Test image	NPCR	UACI
Airplane	99.6284 %	33.4984 %
Baboon	99.6078 %	33.4614 %
Bridge	99.6178 %	33.4485 %
Fishing boat	99.6143 %	33.4596 %
Lena	99.6197 %	33.5282 %
Peppers	99.6040 %	33.4538 %

5. Experimental results

In the experiment, some grayscale images of the size 512×512 were used to evaluate encryption performance. The simulated experiments were performed on an AMD Ryzen 7 5800H CPU with 16G memory. The version of MATLAB used was R2019a. Fig. 13 illustrates the encrypted and decrypted images using the proposed encryption algorithm. Fig. 13 shows that the encrypted images are irregular noise signal images, and it is difficult to observe relevant information from the encrypted images. The decrypted images are consistent with the plain images.

5.1. Histogram analysis

Fig. 14 shows the histogram comparison of the plain images and the encrypted images. It can be seen that the histograms of the encrypted images are completely different from that of the plain images, and the histograms of the encrypted images are almost uniformly distributed. This shows that the encryption algorithm has a good encryption effect and can resist statistical attacks.

5.2. Key space analysis

The total number of possible keys is called the key space. The key space of a secure chaos-based image encryption algorithm should be larger than 2^{100} [20]. The larger the key space of the encryption algorithm, the more resistant ability to brute force attacks. The keys for the proposed algorithm can be the 256-bit hash value H or the three parameters μ , x_0 and y_0 . (1) If the key is the 256-bit hash value H , then the key space is 2^{256} . (2) If the keys are three parameters μ , x_0 and y_0 , then the key space is $10^{14} \times 3 > 2^{142}$. The premise is that the accuracy of the computer is 10^{-14} . Considering that the attacker can attack any key in the above two cases, the final key space should be the minimum value of them. Therefore, the final key space for the proposed algorithm is $10^{14} \times 3$, which is about 2^{142} and larger than 2^{100} .

5.3. Correlation analysis

Encrypted images with low correlation in the horizontal, vertical, and diagonal directions are considered effective in resisting statistical attacks. Fig. 15 shows the results of the correlation of plain images and encrypted images. H, V, and D on the x-axis are the horizontal, vertical, and diagonal directions. The values on the y-axis are the gray values of the tested pixel. The values on the z-axis are the gray values of the adjacent pixels of the tested pixel. Fig. 15 shows that the adjacent pixels of the encrypted images are evenly distributed, which shows that the encryption algorithm eliminates the correlation in all directions of the

image.

Table 3 presents the correlation coefficients of the plain and encrypted images. It can be seen that the correlation coefficients of plain images are all close to 1 and the correlation coefficients of encrypted images are close to 0. Compared with other existing algorithms, the encrypted images of the proposed algorithm have lower correlation coefficients.

5.4. Information entropy analysis

For an image, the more uniform the grayscale distribution, the greater the information entropy. Conversely, the less uniform the grayscale distribution, the smaller the information entropy. The greater the entropy, the less the amount of information. Therefore, the less information is leaked from the plain image. For 256-level grayscale image, the theoretical value of information entropy is 8. Information entropy can describe the confusion extent of an image. Table 4 gives the information entropy values of plain images and encrypted images. Obviously, the information entropy values of the encrypted images are improved compared to the information entropy values of plain images. Table 4 also gives the comparison of the information entropy of the proposed algorithm and other existing algorithms. The information entropy of the proposed algorithm is slightly higher than that of the existing algorithms. Therefore, the proposed algorithm can hide a large amount of information.

5.5. Analysis of differential attack

A differential attack means that the attacker finds the correspondence between the plain image and the encrypted image by comparing the difference between the corresponding ciphertext before and after the plain image is slightly changed. Resistance to differential attacks is typically evaluated using two metrics: Number of Pixels Change Rate (NPCR) and Uniform Average Change Intensity (UACI). The NPCR and UACI values should ideally be about 99.6094 % and 33.4635 %, respectively. Eqs. (16)–(18) is the definitions of NPCR and UACI [26].

$$A(i, j) = \begin{cases} 0, & \text{if } B_1(i, j) = B_2(i, j) \\ 1, & \text{else} \end{cases}, \quad (16)$$

$$NPCR = \frac{\sum_{i=1}^m \sum_{j=1}^n A(i, j)}{m \times n} \times 100\%, \quad (17)$$

$$UACI = \frac{\sum_{i=1}^m \sum_{j=1}^n |B_1(i, j) - B_2(i, j)|}{m \times n \times 255} \times 100\%, \quad (18)$$

where m and n are the height and width of the plain image, and B_1 and B_2 denote the encrypted images with a change in the pixel value and no change in the pixel value. The NPCR and UACI values of different images were calculated, and the experimental results are shown in Table 5. The NPCR and UACI values of proposed algorithm are close to the theoretical value. Table 6 shows the average NPCR and UACI values. Compared with other chaotic image encryption algorithms, the average NPCR and UACI values of the proposed algorithm are closer to the theoretical values than those of the existing algorithms. Therefore, the proposed algorithm can more effectively resist differential attacks.

Table 6

Comparison of the NPCR and UACI.

Metric	Proposed algorithm	Ref. [27]	Ref. [28]	Ref. [29]	Ref. [35]	Ref. [36]
NPCR	99.6153 %	99.6282 %	99.6113 %	99.6086 %	99.6152 %	99.6078 %
UACI	33.4750 %	33.2459 %	33.4708 %	33.4635 %	33.4763 %	33.5708 %

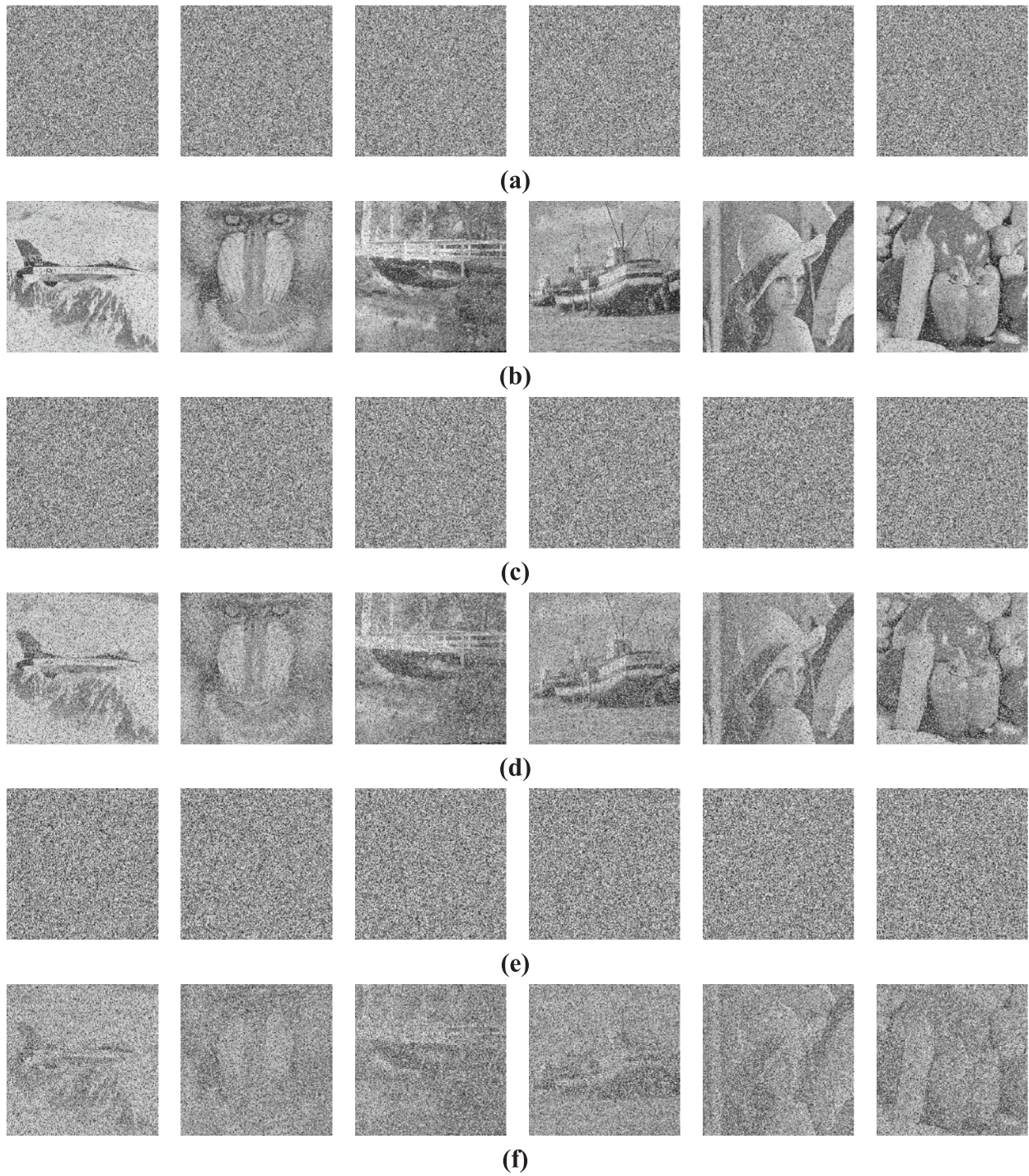


Fig. 16. Analysis of salt and pepper noise. (Left to right: Airplane, Baboon, Bridge, Fishing boat, Lena and Peppers.) (a) encrypted images by accumulating pepper and salt noise with noise density 0.05, (b) the decrypted images obtained from (a), (c) encrypted images by accumulating pepper and salt noise with noise density 0.1, (d) the decrypted images obtained from (c), (e) encrypted images by accumulating pepper and salt noise with noise density 0.2, (f) the decrypted image obtained from (e).

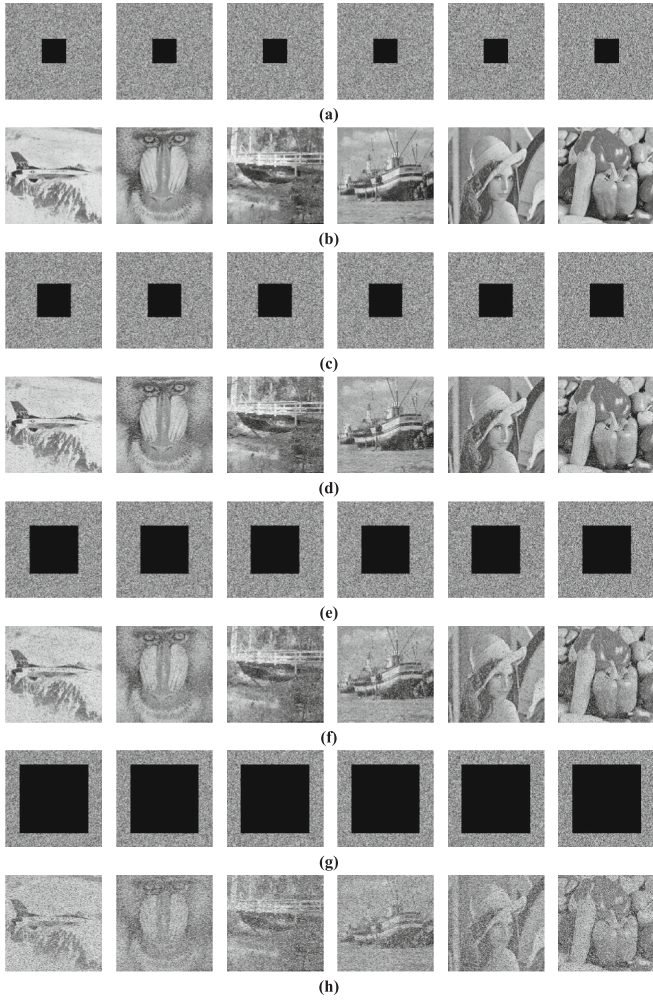


Fig. 17. Analysis of clipping attacks. (left to right: Airplane, Baboon, Bridge, Fishing boat, Lena and Peppers.) (a) 1/16 data loss in the encrypted images, (b) the decrypted image obtained from (a), (c) 1/8 data loss in the encrypted images, (d) the decrypted image obtained from (c), (e) 1/4 data loss in the encrypted images, (f) the decrypted image obtained from (e), (g) 1/2 data loss in the encrypted images, (h) the decrypted image obtained from (g).

Table 7
PSNR and MSE of the salt and pepper noise attack.

Test image	Metric	0.05	0.1	0.2
Airplane	PSNR	13.8528	11.3259	9.4019
	MSE	2677.9608	4791.6768	7462.6027
Baboon	PSNR	15.2874	12.8137	10.8775
	MSE	1924.5786	3401.7852	5312.8276
Bridge	PSNR	14.5323	12.0589	10.1086
	MSE	2290.0716	4047.5731	6341.9272
Fishing boat	PSNR	14.9673	12.5192	10.5983
	MSE	2071.8338	3640.4541	5665.6352
Lena	PSNR	14.8592	12.4343	10.5450
	MSE	2124.0528	3712.3536	5735.5641
Peppers	PSNR	14.5252	12.0759	10.1703
	MSE	2293.8225	4031.6901	6252.4238

5.6. Cropping and noise attacks analysis

In a real transmission environment, image transmission will be at risk of noise. A good encryption algorithm is expected to successfully cope with them. Fig. 16 shows the resistance ability to the salt and pepper noise. The pepper and salt noises were mixed with various densities of 0.05, 0.1 and 0.2 in the encrypted images of Lena.

Table 8
PSNR and MSE of clipping attacks.

Test image	Metric	1/16	1/8	1/4	1/2
Airplane	PSNR	20.1585	17.3477	14.3198	11.3486
	MSE	626.9448	1197.5942	2404.9205	4766.7724
Baboon	PSNR	21.1585	18.5659	15.5567	12.5454
	MSE	470.7813	904.6657	1808.9008	3618.5667
Bridge	PSNR	20.4369	17.5768	14.5678	11.5706
	MSE	588.0124	1136.0631	2271.4401	4529.1763
Fishing boat	PSNR	21.1268	18.5659	15.2181	12.2493
	MSE	501.6494	975.5815	1955.5394	3873.9131
Lena	PSNR	21.0070	18.1899	15.1315	12.1538
	MSE	515.6768	986.4857	1994.9501	3960.0381
Peppers	PSNR	20.6172	17.7608	14.7380	11.7273
	MSE	564.1092	1088.9390	2184.1253	4368.6467

Apparently, these decrypted images are still identifiable, which shows that the proposed algorithm is resistant to noise attacks.

To test the algorithm resistance to the cropping attack, different sizes of cropping are performed at different locations of the encrypted image. Fig. 17 shows the encryption resistance ability to clip attacks. Fig. 17(a)–(d) represents the encrypted images with 1/16, 1/8, 1/4 and 1/2 data loss in the encrypted images of Lena respectively. The plain image is easily recognized, which means that the proposed encryption algorithm can prevent data loss attacks.

Peak signal-to-noise ratio (PSNR) can evaluate the quality of the decrypted image. The larger the PSNR value, the better the quality of the decrypted image. That means the decrypted image is closer to the plain image. PSNR can be calculated based on the mean square error (MSE) between the plain image and the decrypted image. Eqs. (19) and (20) are the definitions of MSE and PSNR.

$$MSE = \frac{\sum_{i=1}^m \sum_{j=1}^n (P(i,j) - D(i,j))^2}{m \times n}, \quad (19)$$

$$PSNR = 10 \times \log_{10} \left(\frac{255^2}{MSE} \right), \quad (20)$$

where P represents the plain image and D represents the decrypted image. Tables 7–8 give the PSNR and MSE of salt and pepper noise attack and cropping attack respectively. Although, the PSNR values are decreased with the intensity of cropping attack and salt and pepper noise attack, the PSNR values are maintained at a good value.

5.7. Chosen plaintext attack analysis

The chosen plaintext attacks are common attacks in the field of image encryption. The attacker can select some plaintext and obtain the corresponding ciphertext. Use the known plaintext-ciphertext pairs to crack the encryption algorithm's key or obtain information about other unknown plaintext. To verify the ability of the encryption algorithm to resist chosen plaintext attacks, six pure black and pure white images were encrypted respectively. Fig. 18 shows the encryption effect. It can be seen that the encrypted images of the two special images are similar to noise images. Table 9 lists the information entropy and correlation coefficients of the two special images. The encrypted images are similar to the noise, the histograms are flat, the values of information entropy are all greater than or equal to 7.9993, and the correlation coefficients are all close to 0. This shows that the proposed encryption algorithm can effectively resist chosen plaintext attacks.

5.8. Encryption time analysis

The encryption computational efficiency mainly depends on the characteristics, implementation methods, and computational resources of the selected encryption algorithm. Under the same experimental conditions, such as hardware, software platforms and encryption

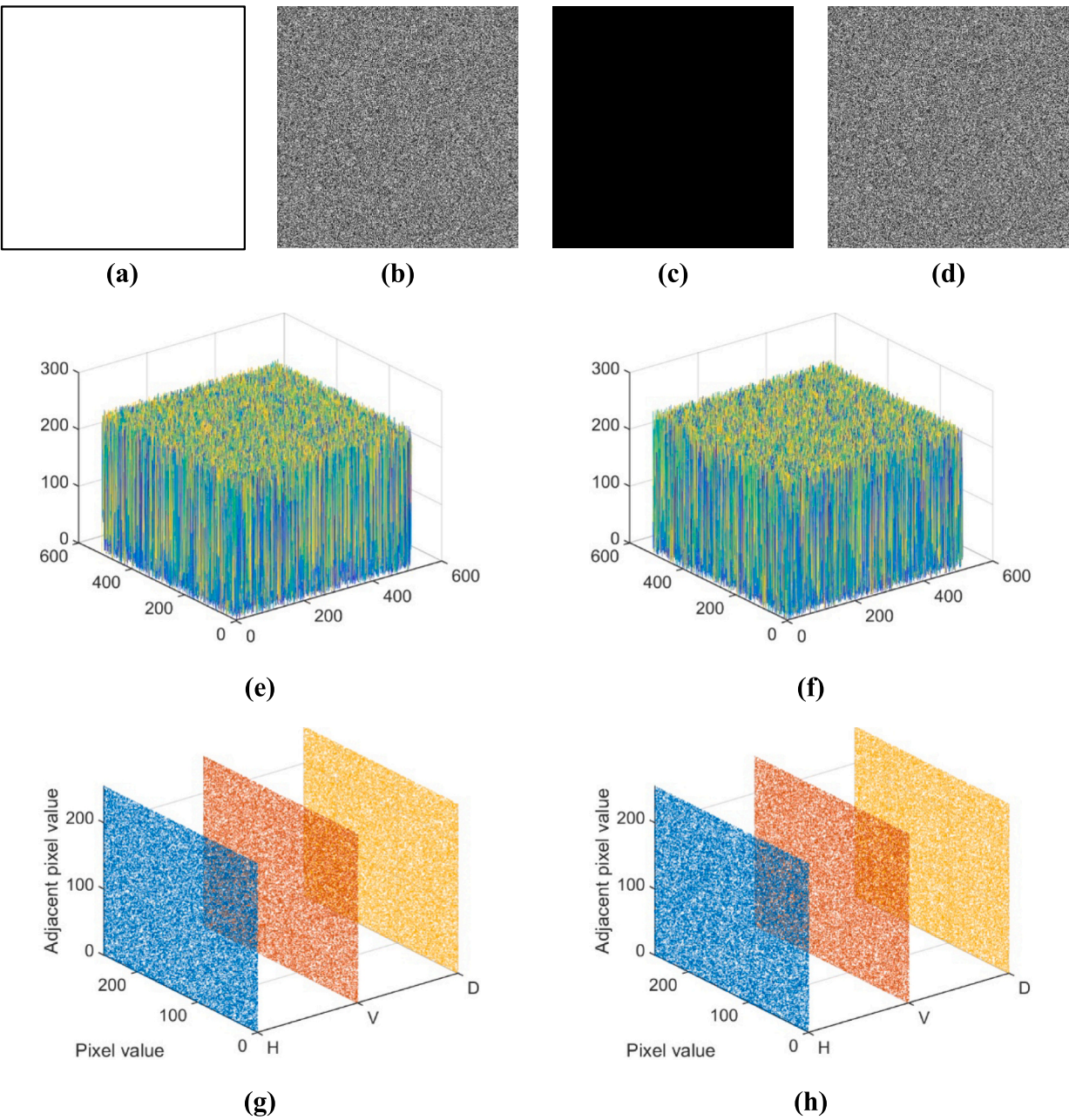


Fig. 18. Encryption result of White and Black images. (a) White image, (b) encrypted White, (c) Black image, (d) encrypted Black, (e) histogram of encrypted White, (f) histogram of encrypted Black, (g) correlation of encrypted White, (h) correlation of encrypted Black.

Table 9
Information entropy and correlation coefficients for White and Black images.

Test image	Information entropy	Correlation coefficients		
		Horizontal	Vertical	Diagonal
White	7.9993	0.0060	0.0060	0.0068
Black	7.9993	0.0032	−0.0066	0.0007

Table 10
Comparison of encryption times for different algorithms (second).

Algorithm	Hardware	Software	Image size	Encryption time
Proposed algorithm	3.20 GHz CPU, 16G RAM	MATLAB R2019a	512 × 512	0.9245
Ref. [19]	3.20GHz CPU, 16G RAM	MATLAB R2021b	512 × 512	2.0915
Ref. [24]	3.20 GHz CPU, 16G RAM	MATLAB R2019a	512 × 512	2.8251
Ref. [15]	3.20 GHz CPU, 16G RAM	–	512 × 512	2.0866

objects, an encryption algorithm that encrypts the same amount of data but takes less time to encrypt has higher encryption computational efficiency. In the test, six grayscale images of the size 512 × 512 were used as the plain images. The average encryption time of one image is calculated, and the results are listed in Table 10. Compared with other recently proposed encryption algorithms, the proposed algorithm takes less time to encrypt a 512 × 512 size image under the same hardware environment. Test results show that the proposed algorithm is efficient.

6. Conclusions

This paper designs a composite 2D chaotic map named the 2DSHM, and tests show that it has good randomness. Subsequently, a new multiple-image encryption algorithm is proposed based on the designed chaotic map. The proposed encryption algorithm uses the SHA-256 algorithm to generate the initial values of designed chaotic map. Then, the designed chaotic map is used to generate a 3D coordinate matrix and change the pixels position. The image diffusion is achieved by combining chaos and the XOR operation. Experimental results show that the proposed encryption algorithm can hide plaintext information, resist various noise attacks, clipping attacks, etc.

CRediT authorship contribution statement

Qisu Gao: Writing – review & editing, Writing – original draft, Methodology. **Xiaoqiang Zhang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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